TriAlign Engineering Plan

System Design for Clinically Meaningful Eligibility-Criterion Retrieval and Comparison

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1 Read This First

Before starting, each team member should read the original AutoCriteria paper [1] and understand its basic workflow: an LLM reads free-text eligibility criteria from a clinical trial and converts them into a more structured format. The paper is the starting point for this project.

Team members should then review the Modern AutoCrit repository [2]:

`https://github.com/JonathanMa03/modern-autocrit/tree/main`

The repository contains the working system that this project will extend. Teammates do not need to understand every implementation detail before contributing, but they should be able to run the application, inspect one completed extraction, and follow a criterion from source protocol text to its structured output.

Modern AutoCrit is our expanded implementation of that idea. The original AutoCriteria work established the protocol-to-structured-criteria concept. Modern AutoCrit adds the project infrastructure needed for TriAlign:

- retrieval of ClinicalTrials.gov records by NCT identifier;
- parsing of ClinicalTrials.gov API v2 records;
- atomic segmentation of compound eligibility text into separately comparable criteria;
- Chia-inspired structured criteria with domain, entity, comparator, value, unit, interval, negation, timing, repetition, qualifiers, and Boolean scope;
- stable criterion and parent-statement identifiers, source versions, surrounding context, and protocol provenance;
- governed clinical terminology and canonical concepts;
- separate extraction confidence, automated validation findings, and human-review status;
- recoverable single-trial and manifest jobs with persistent batch checkpoints;
- a frozen, versioned export containing only reviewed criteria that pass TriAlign-readiness checks;
- modular backend code rather than one long extraction script; and
- a browser interface for running extraction and inspecting results.

The features most relevant to TriAlign are atomic segmentation, normalized structured values, stable identifiers, Boolean scope, surrounding context, terminology normalization, source evidence, extraction uncertainty, automated validation, and reviewer decisions. They provide the inputs and quality signals needed to construct criterion pairs. The browser interface and job-recovery layer are useful engineering support, but they are not the central contribution of TriAlign.

You do not need to understand the entire codebase immediately. The important input and output are:

Protocol eligibility text $\longrightarrow$ atomic, structured eligibility criteria

For example, Modern AutoCrit should turn "participants must be at least 18 years old" into a record containing the concept *age*, the operator $\geq$, the value 18, and the unit years.

This project begins after extraction. We want to compare criteria from different protocols and train an embedding model that understands clinically important differences.

2 Product Goal

Current sentence-embedding models often reward similar wording. That is not always clinically correct:

- "age ≥ 18" and "age < 18" look almost identical but describe different groups;
- "no cardiovascular disease" and "controlled hypertension is allowed" discuss the same topic but are not equivalent;
- "creatinine clearance ≥ 30 mL/min" and "eGFR ≥ 30 mL/min/1.73 m^2" may look similar but use different measurements; and
- two differently worded criteria may actually mean the same thing.

Our goal is to build a new dense embedding model for eligibility criteria. Criteria with similar clinical meaning should be close in the learned space. Criteria that only look similar, but differ because of negation, operators, thresholds, timing, or exceptions, should be separated.

The system will be a usable clinical data-science tool rather than only a model-development exercise. A user should be able to supply or select a criterion, retrieve comparable criteria from other protocols, inspect their structured differences, see an interpretable relationship and uncertainty estimate, and review the original evidence. The system will use one agent to find difficult examples, choose comparison tools, ask an LLM for an initial judgment when clinical context is needed, run exact checks when the relation is computable, send uncertain cases to humans, and turn accepted labels into model-training data.

One-sentence aim

Build and evaluate an agentic workflow that creates human-validated criterion pairs from clinical protocols and uses them to fine-tune a clinically meaningful dense embedding model.

Primary project question

Where do generic dense embeddings fail to capture clinically meaningful differences in trial eligibility criteria, and can an agent combine LLM judgments, structured checks, and human review to fine-tune an embedding model that handles those differences?

Three research questions

1. Which clinical phenomena—including negation, operators, thresholds, units, timing, prior treatment, exceptions, and compound logic—cause conventional embeddings to produce clinically misleading rankings?
2. Does fine-tuning one open dense-encoder checkpoint on human-validated criterion pairs and hard negatives improve equivalence retrieval and anchor ordering over frozen lexical, general, and biomedical baselines?
3. Does the agentic workflow improve labeling quality, calibration, and reviewer efficiency while appropriately abstaining or escalating uncertain cases?

3 High-Level Architecture

TriAlign separates reviewed source data, offline model development, and the runtime comparison application. Model weights never change during a user request; new labels enter a versioned offline training run only after review.

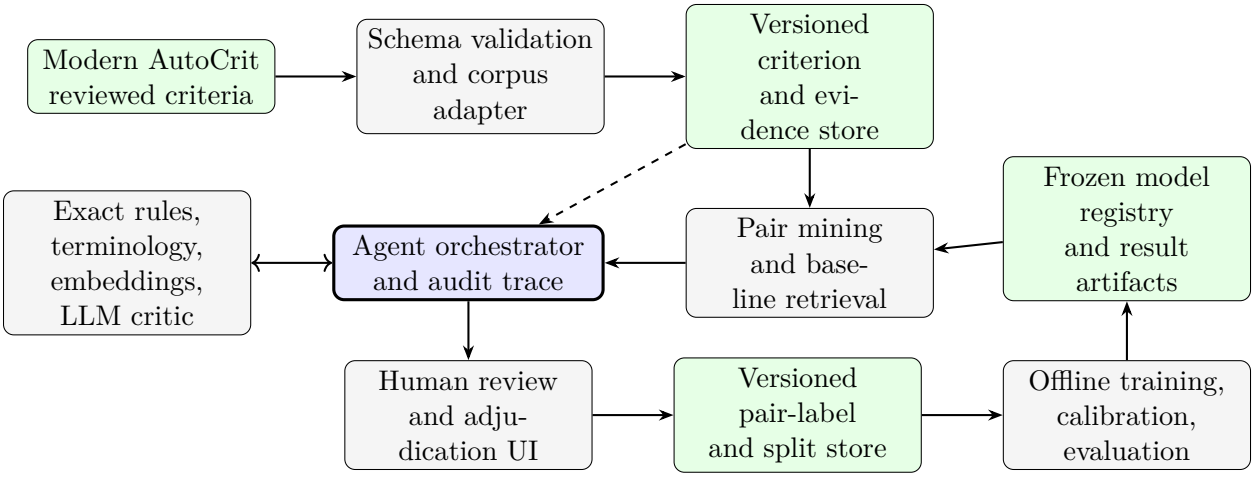


Figure 1: Planned TriAlign data, labeling, training, and serving flow.

3.1 System boundaries

Subsystem	Responsibility	Primary artifact
Corpus adapter	Read the Modern AutoCrit workbook, validate required fields, normalize types, and preserve immutable source evidence.	Versioned Parquet/JSON criterion table plus validation report.
Pair miner	Build random controls, nearest-neighbor candidates, structured challenge pairs, and trial-family-safe partitions.	Pair manifest with sampling provenance and split identifiers.
Agent runtime	Select tools, compare evidence, request an LLM judgment when needed, check conflicts, estimate uncertainty, and escalate.	Append-only trace containing inputs, tool versions, outputs, and routing decision.
Review application	Show both criteria and context, collect accept/correct/ambiguous decisions, and support adjudication.	Versioned human label and rationale; no silent overwrite.
Training pipeline	Fine-tune the encoder, fit the relationship model, calibrate probabilities, and run seeded evaluation.	Frozen model bundle, configuration, metrics, and model card.

Retrieval service	Embed a query with a frozen model, retrieve candidates, apply relationship prediction, and return evidence.	Ranked results with model version, relationship, uncertainty, and provenance.

3.2 Repository layout to implement

```
trialign/
  configs/              # versioned data, model, prompt, and experiment configs
  data/                 # raw, interim, reviewed, split, and pair manifests
  src/trialign/
    data/               # workbook adapters, schemas, validation, split logic
    pairing/            # candidate mining and controlled challenge generation
    tools/              # interval, unit, negation, terminology, and logic checks
    agents/             # orchestration, critic, routing, uncertainty, audit trace
    models/             # embedding, relationship classifier, calibration, registry
    evaluation/         # retrieval, classification, calibration, and statistics
    api/                # runtime service interfaces
  app/                  # human review and comparison interface
  tests/                # unit, schema, leakage, integration, and regression tests
  scripts/              # reproducible CLI entry points
  artifacts/            # ignored local run outputs; released artifacts are versioned
```

The first implementation should remain modular but small. A Python state machine with typed tool interfaces is preferred over adopting an agent framework before the workflow stabilizes. Every stage must also be runnable from a command line so experiments do not depend on the web interface.

3.3 Dashboard and user workflow

The dashboard is the operational interface to TriAlign, not a collection of decorative summary charts. It turns retrieval, relationship prediction, agent routing, source evidence, uncertainty, and human steering into one reviewable workflow. A clinical data scientist should be able to:

1. search or filter the reviewed corpus by trial, therapeutic area, criterion type, domain, entity, relationship, uncertainty, or review status;
2. select one criterion and retrieve comparable criteria from other protocols with the frozen encoder;
3. inspect the two source passages, parent statements, structured fields, and protocol links side by side;
4. see embedding similarity separately from the predicted clinical relationship and its direction;
5. inspect the agent's selected tools, exact-check results, evidence, disagreements, calibrated uncertainty, and escalation reason;
6. accept, correct, reject, or mark the proposed relationship as ambiguous; and
7. export a versioned decision with reviewer rationale and a complete audit reference.

The initial dashboard should have five compact views:

- **Search:** select a trial or individual reviewed criterion and apply corpus filters.

- **Ranked comparisons:** display cross-trial candidates, embedding scores, predicted relationships, uncertainty, and review state.
- **Comparison detail:** show source evidence and field-level differences for polarity, entity, comparator, value, unit, interval, negation, timing, repetition, exceptions, and Boolean scope.
- **Review queue:** prioritize tool disagreements, input-order inconsistencies, unfamiliar concepts, high uncertainty, and explicit agent abstentions.
- **Evaluation:** summarize retrieval, relationship classification, calibration, risk–coverage, reviewer corrections, latency, and cost by model version and clinical phenomenon.

The principal product output is a ranked, evidence-linked comparison rather than a single similarity score. Each result must distinguish the symmetric embedding score from the explicit relationship label, such as equivalent, more restrictive, partially overlapping, disjoint, unrelated, or ambiguous. It must also retain both criterion identifiers, trial and protocol provenance, model and prompt versions, tool traces, uncertainty, escalation status, and any human correction.

Training, corpus freezing, pair generation, and bulk evaluation remain command-line workflows. The dashboard calls the same importable, versioned application services and must not contain separate model logic. Runtime requests use frozen model weights; accepted dashboard decisions enter a new versioned offline training set only after validation. Streamlit is sufficient for the first prototype, while the service boundary should permit a later FastAPI-backed interface without rewriting comparison logic.

3.4 Core records and interfaces

The minimum stable records are:

- `CriterionRecord` and `PairCandidate`;
- `AgentJudgment` and `HumanDecision`; and
- `DatasetSplit` and `ModelVersion`.

Each record should include a schema version, stable identifier, creation time, parent artifact identifiers, and code/model/prompt version where applicable. Pydantic models and exported JSON Schemas should enforce these interfaces. Migrations must create a new artifact rather than modifying reviewed data in place.

4 Task Definition

For each pair of criteria, we will assign one of the following relationships:

- equivalent;
- partially overlapping or compatible;
- criterion A is more restrictive;
- criterion B is more restrictive;
- contradictory or effectively disjoint;
- related topic, but not directly comparable;
- unrelated; or

- too ambiguous to decide.

The main model has two jobs:

1. **Retrieval:** given one criterion, retrieve other criteria with comparable clinical meaning.
2. **Relationship prediction:** explain whether a retrieved pair is equivalent, overlapping, directional, contradictory, or unrelated.

These are separate because ordinary embedding distance is symmetric, while "A is more restrictive than B" is directional.

The annotation guide will be reviewed with the instructor before the benchmark is frozen. It will define each label, give positive and counterexamples, require reviewers to interpret inclusion/exclusion polarity before judging restrictiveness, and specify when evidence is insufficient. Every label will retain the source evidence and a concise rationale.

5 Data Pipeline and Benchmark Construction

5.1 Protocol corpus

Clinical protocols are the main source material. The target corpus is 80–120 public trials and approximately 2,000 atomic criteria from one or two therapeutic areas. Oncology and cardiovascular disease are the leading application areas because they provide complementary challenges: oncology emphasizes biomarkers, stage, performance status, prior therapy, and nested exceptions, while cardiovascular protocols combine measurements, event history, treatment stability, and temporal restrictions. If complete review at the target size is infeasible, we will report the achieved corpus size rather than silently weakening review requirements. For every protocol, we will keep:

- the original eligibility text;
- individual criteria extracted by Modern AutoCrit;
- structured values, operators, units, negation, timing, and other qualifiers;
- stable protocol, parent-statement, and atomic-criterion identifiers;
- the source protocol version, location, surrounding text, inclusion/exclusion context, and reuse conditions;
- extraction confidence and human-review status as separate variables; and
- basic trial metadata.

The dataset must include more than age. We want examples involving demographics, laboratory thresholds, disease stage, biomarkers, prior treatment, medication use, comorbidities, performance status, reproductive requirements, timing, severity, negation, exceptions, and AND/OR scope. Laboratory, disease, and treatment criteria must all be represented.

Chia is a useful example of a large structured eligibility-criteria dataset [3]. Prior work has also studied embeddings for eligibility clustering and trial similarity [4, 5]. We still need our own pair labels because those resources do not directly answer our relationship question.

5.2 How we find useful pairs

First, we will embed all criteria using three frozen general encoders and, where feasible, one biomedical encoder. Development examples will be mined across multiple encoders rather than selected

from the errors of only one model. We will record the encoder, query, rank, sampling rule, and reason each pair entered the dataset. We will then inspect nearest neighbors and deliberately collect cases where conventional embeddings fail. Useful cases include:

- opposite operators or negation despite high similarity;
- changed values, unit conversions, incompatible measurements, or time windows;
- equivalent meaning written differently;
- broad rules paired with narrower exceptions or nested requirements;
- similar vocabulary used for different clinical purposes; and
- uncertain or poorly extracted criteria.

Natural cross-trial comparisons and constructed challenges will be marked and analyzed separately. Constructed challenges may include controlled paraphrases, converted units, changed thresholds, reversed operators, inserted or removed negation, nested requirements, and unrelated controls. Anchor triplets will encode expected ordering; for example, "age at least 18" should be closer to "age 18 or older" than to "age below 18." We will also include random and clearly unrelated pairs. Otherwise, a benchmark made only from difficult failures would not reflect normal model behavior. Random sampling will later serve as a control for uncertainty-based selection.

Training, validation, and test sets will be separated by trial family *before* criterion pairs are constructed. Amendments, shared protocol templates, near duplicates, and constructed examples derived from them must stay in the same partition. We will record provenance and permitted reuse conditions, keep final test labels inaccessible to agent retrieval, and report possible encoder pretraining exposure to public trial text. Pilot and test comparisons will be independently double labeled, with qualified clinical adjudication for unresolved clinically material disagreements. During initial test review, reviewers will not see model scores or LLM proposals.

6 Labeling Pipeline

The workflow is:

Protocol pair → agent tools → LLM suggestion → human review → accepted label → model training.

6.1 LLM suggestion

The LLM receives the two criteria, their structured fields, and enough versioned protocol context to interpret qualifiers and exclusion logic. It proposes a relationship, the clinically important difference, supporting evidence, confidence, uncertainty, and a short rationale. This proposal is not treated as truth and LLM agreement is never the reference standard.

Automatic checks reject obvious mistakes. For example, a label should reverse correctly when A and B are swapped, incompatible units should be flagged, and exact disjoint intervals cannot be called overlapping.

6.2 Human review and steering

Reviewers accept, correct, reject, or mark each proposed label as ambiguous. Pilot and test examples will be independently reviewed by two people so that we can improve the label guide and

measure agreement. Clinically consequential disagreements will receive qualified adjudication. Only accepted or adjudicated labels enter core training; unresolved cases remain visible rather than being forced into a class.

As the model improves, the agent will prioritize examples where:

- the model is uncertain;
- the LLM and structured tools disagree;
- input order changes the answer;
- the example is unlike the existing training data; or
- reviewers frequently disagree.

This is the human-steering loop: human corrections guide which examples are collected next and how the model is retrained.

To evaluate whether targeted selection genuinely helps, uncertainty/disagreement sampling will be compared with random sampling under the same review budget. Manual annotation, LLM-assisted review, and the complete agent workflow will also be compared on matched tasks rather than on different examples.

7 Model Development Plan

We will start from one open pretrained text-encoder checkpoint rather than train a language model from scratch. We will compare it with three frozen generic encoders and a biomedical encoder where feasible, then fine-tune the selected open checkpoint on human-validated pairs. The contribution is the new protocol-derived dataset, clinically difficult negatives, training setup, agentic labeling process, and evaluation rather than merely calling an existing Hugging Face model.

7.1 Candidate checkpoints and comparison tools

The final checkpoint is a validation-set decision, not a commitment made in this document. Every candidate must have an acceptable license, a reproducible model revision, and an input length suitable for atomic criteria plus limited context.

Role	**Candidate**	**Tool or source**
Lexical baseline	TF–IDF and cosine similarity	Scikit-learn `TfidfVectorizer`: `https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html`.
Small general encoder	`all-MiniLM-L6-v2`	Hugging Face model: `https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2`. Useful fast baseline, not assumed to be clinically adequate.
Strong general encoder	`all-mpnet-base-v2`	Hugging Face model: `https://huggingface.co/sentence-transformers/all-mpnet-base-v2`.

Role	Candidate	Tool or source
Retrieval-oriented encoder	`bge-base-en-v1.5`	Hugging Face model: `https://huggingface.co/BAAI/bge-base-en-v1.5`. Input formatting and recommended query instructions must be recorded.
Biomedical encoder	BioLORD-2023	Hugging Face model: `https://huggingface.co/FremyCompany/BioLORD-2023`. Evaluate whether its biomedical concept training transfers to full eligibility statements.
Biomedical concept baseline	SapBERT	Hugging Face model: `https://huggingface.co/cambridgeltl/SapBERT-from-PubMedBERT-fulltext`. This is primarily a biomedical entity baseline, not automatically a sentence-retrieval model.
Embedding training	Sentence Transformers	Training and evaluator APIs: `https://www.sbert.net/docs/sentence_transformer/training_overview.html`. Candidate losses include multiple-negatives ranking, triplet, contrastive, or CoSENT losses, selected to match the final label format.
Model and data access	Hugging Face Hub, Transformers, Datasets	Pin repository revisions and package versions; cache downloaded artifacts; store licenses and model cards with experiment metadata.
Vector retrieval	Exact NumPy/scikit-learn search first; FAISS if needed	FAISS: `https://github.com/facebookresearch/faiss`. With only a few thousand criteria, exact normalized dot-product search should remain the reference implementation.
Relationship model	Scikit-learn or PyTorch classifier	Start with logistic regression or a small multilayer head over $[u, v, \lvert u - v \rvert, u \odot v]$ before considering a cross-encoder.
Experiment tracking	Local JSON/Parquet plus MLflow or Weights & Biases if useful	The required interface is a reproducible local run manifest; a hosted tracker is optional.
Statistics	NumPy, SciPy, scikit-learn, statsmodels	Bootstrap by trial family, paired tests, calibration curves, confusion matrices, and risk–coverage analysis.
Application/API	FastAPI plus a small review UI, or Streamlit for the first prototype	Choose after the review interaction is sketched; all business logic must stay in importable modules and CLI commands.

Role	Candidate	Tool or source
Agent/LLM layer	Explicit Python orchestrator plus provider adapter	Keep tool calls typed and auditable. Direct provider SDKs are preferred initially; an agent framework is optional rather than an architectural dependency.

Before downloading or training a model, record its exact Hub identifier, revision hash, license, pooling method, maximum sequence length, embedding dimension, input prefix requirements, and normalization convention. The baseline runner should expose one common `encode(texts, config)` interface and save embeddings with a corpus fingerprint so results cannot be mixed across data versions.

Training examples have three broad roles:

- **Positive pairs:** equivalent or clinically interchangeable criteria;
- **ordinary negatives:** unrelated criteria; and
- **hard negatives:** criteria with similar wording but an important clinical difference.

The encoder $f_\theta(c)$ maps criterion c to a vector. Its basic distance is

$$d_\theta(A, B) = 1 - \cos\{f_\theta(A), f_\theta(B)\}.$$

Contrastive or triplet training pulls positive pairs closer and pushes appropriate negatives apart. Hard negatives teach the model that an operator, number, negation, time window, or exception can matter even when the rest of the sentence is unchanged. Partially overlapping requirements should not automatically be pushed as far apart as unrelated criteria; label-aware sampling or loss weighting must preserve that distinction.

A small classification head will use both criterion vectors to predict the explicit relationship label. This head handles directional relationships that cosine distance alone cannot represent.

Structured rules are used for labeling and quality control. We will not average structured, embedding, population, and LLM scores into one score with unclear clinical meaning.

Fine-tuning occurs offline. Runtime comparisons use frozen, versioned weights. The application will show distances, nearest neighbors, source evidence, structured differences, predicted relationships, uncertainty, and review decisions rather than silently updating the model during use.

7.2 Minimum training experiment

The first trainable experiment should use one selected Sentence Transformers checkpoint, a frozen trial-family split, one positive/negative pair format, one hard-negative strategy, and at least three random seeds. It must compare the fine-tuned model with the unchanged starting checkpoint on the same judged candidate pool. Training should stop on a validation retrieval metric; the test set is evaluated once after choices are frozen. Larger sweeps, parameter-efficient tuning, cross-encoders, or multi-task losses are follow-up experiments rather than prerequisites.

8 Population Information from Manuscripts

We probably will not have individual patient data. Therefore, we will not claim to calculate how many individual patients would change eligibility between two trials, and we will not simulate an unknown patient population as if it were observed.

Instead, we may link protocols to their trial manuscripts and collect reported baseline characteristics from Table 1 or supplementary tables. Examples include age, sex, race or ethnicity, disease stage, performance status, prior treatment, and disease-specific severity measures.

For a characteristic reported by both trials, we can calculate a standardized difference. For continuous variables:

$$\Delta_k(a, b) = \frac{\bar{x}_{ak} - \bar{x}_{bk}}{\sqrt{(s_{ak}^2 + s_{bk}^2)/2}}.$$

Binary proportions can be standardized in a similar way. The main output should be a table or profile showing which reported characteristics differ, not a single unexplained population score.

This is an optional secondary descriptive analysis, not part of the primary embedding benchmark. It asks whether protocols that look similar according to their eligibility criteria also report similar enrolled baseline characteristics. It does not analyze population outcomes, determine individual patient eligibility, establish causality or transportability, or quantify clinical risk. Published summaries omit information and cannot recover the full joint patient distribution. Population-adjustment literature explains these limitations [13, 14].

Before implementing this analysis, we need a short literature review and a common dictionary for baseline variables. If too few manuscripts report comparable fields, this component should be reduced rather than filled with unsupported assumptions.

9 Agent Runtime Design

One agent coordinates a workflow rather than simply generating an answer. Its tools cover Modern AutoCrit extraction, retrieval, frozen embeddings, LLM judgment, exact rule checking, and human review. The system preserves inputs, model and prompt versions, intermediate judgments, corrections, uncertainty, and final decisions. It will:

1. confirm protocol versions and reviewed criterion interpretations;
2. run frozen encoders and organize representative and difficult comparisons by concept, clinical phenomenon, and sampling source;
3. select candidate criterion pairs and retrieve original context without losing qualifiers or exclusion logic;
4. choose structured, terminology, embedding, or LLM tools;
5. request an LLM relationship, clinical difference, evidence, rationale, and uncertainty;
6. check units, intervals, operators, negation, and label reversal when A and B are swapped;
7. send uncertain cases to human reviewers;
8. record the final label and its provenance;
9. choose new examples using uncertainty and disagreement while retaining random sampling as a control; and
10. apply frozen, versioned weights and display distances, neighbors, evidence, and review decisions with an audit trail.

The agent does not change model weights during a comparison and does not invent a distance. Training happens offline on reviewed data. At runtime, the agent loads a versioned model and decides whether a result is reliable enough to accept.

10 Evaluation Implementation

10.1 Models to compare

- scikit-learn TF–IDF with cosine similarity as the transparent lexical baseline;
- frozen `all-MiniLM-L6-v2`, `all-mpnet-base-v2`, and `bge-base-en-v1.5` general/retrieval encoders;
- frozen BioLORD-2023 and, if useful for entity-focused cases, SapBERT biomedical baselines;
- structured rules where applicable;
- LLM judgment without fine-tuning;
- our fine-tuned dense encoder; and
- the full agent workflow with review and abstention.

Fine-tuning will be compared with its own frozen starting checkpoint and the strongest baseline selected using validation data. Supervision experiments will reuse the same encoder, splits, and review budget. We will also remove important pieces one at a time, such as hard negatives, LLM suggestions, human corrections, targeted sampling, or critic review, to see what actually improves performance and whether fine-tuning degrades ordinary paraphrase recognition.

10.2 Measures

For retrieval, we will report anchor-ordering accuracy, equivalence-retrieval precision and recall@k, mean reciprocal rank, and ranking quality. At a threshold selected only on validation data, we will report both the fraction of non-equivalent pairs incorrectly accepted and the fraction of true equivalents recovered, overall and by clinical phenomenon. For relationship prediction, we will report macro-F1, class-level errors, confusion matrices, probability calibration, abstention, and input-order consistency.

LLM suggestions will be compared with independently adjudicated labels for relationship accuracy, evidence support, input-order consistency, abstention, and reviewer corrections. Unsupported judgments and clinically meaningful failures will be analyzed explicitly.

All systems will receive identical inputs with documented formatting and will be evaluated over fully judged candidate pools. Natural and constructed comparisons will be reported separately. Paired uncertainty estimates will resample disjoint trial-family groups rather than individual pairs because comparisons from one family are dependent; training-seed variation will also be reported.

For the agent, manual annotation, LLM-assisted review, and agent-assisted review will be tested on matched tasks and budgets. We will measure acceptance and correction, conflict detection, reviewer time, escalation errors, latency, monetary cost, and performance change after each annotation round.

Age remains a useful unit test for boundaries and operators. It is not the main benchmark.

11 Primary Scope and Explicit Exclusions

The primary scope is similarity ranking and equivalence retrieval for individual atomic criteria, plus a separate relationship classifier. Shared clinical topic is not treated as equivalent eligibility meaning. Cosine distance will not be presented as a directional restrictiveness score or a measure of clinical risk.

The system is decision support for protocol comparison, not an automated patient-screening device. Individual patient eligibility decisions, patient-level outcome modeling, causal transportability claims, and a single weighted hybrid score with ambiguous clinical meaning are outside scope. The optional manuscript analysis remains descriptive and contingent on comparable aggregate reporting.

12 Implementation Workstreams

These are tasks to complete, not permanent assignments to named team members. Ownership can change by milestone, but every pull request should identify the workstream, acceptance criteria, reviewer, and affected artifact versions.

Data and benchmark

- ☐ Define and test the Modern AutoCrit workbook adapter and stable TriAlign schemas.
- ☐ Complete criterion review, publish correction rules, and freeze a versioned source corpus.
- ☐ Define trial-family groups before pair generation and add automated leakage tests.
- ☐ Write the relationship-label guide with counterexamples and an explicit ambiguous option.
- ☐ Implement random, nearest-neighbor, structured, and controlled-challenge pair samplers with provenance.
- ☐ Build adjudication exports and calculate reviewer agreement by label and clinical phenomenon.

Baselines, training, and statistics

- ☐ Implement one configuration-driven runner for every lexical and embedding baseline.
- ☐ Cache versioned embeddings and build exact-search retrieval before adding an approximate index.
- ☐ Produce the embedding-gap report and select one checkpoint for fine-tuning using validation data.
- ☐ Implement pair/triplet datasets, hard-negative sampling, seeded training, checkpointing, and model cards.
- ☐ Build the directional relationship classifier and probability-calibration step.
- ☐ Implement retrieval, classification, calibration, risk–coverage, bootstrap, and ablation reports.

Agent, review application, and integration

- ☐ Define typed tool contracts for retrieval, structured comparison, terminology, LLM judgment, critic checks, and escalation.

☐ Implement an explicit state machine and append-only audit trace with resumable jobs.

☐ Create the dashboard with criterion search, ranked cross-trial retrieval, side-by-side source context and field differences, agent trace, uncertainty, correction controls, and export.

☐ Add a review queue for disagreement, order sensitivity, out-of-distribution inputs, abstention, and unresolved ambiguity.

☐ Add an evaluation view for retrieval, relationships, calibration, risk–coverage, reviewer corrections, latency, and cost without duplicating experiment logic in the UI.

☐ Add consistency checks for swapped input order, intervals, units, negation, and Boolean scope.

☐ Implement model-registry loading, runtime retrieval, abstention, and evidence-linked result display.

☐ Add unit, integration, regression, and end-to-end tests plus one-command developer workflows.

Shared completion work

☐ Review difficult clinical examples together and adjudicate clinically material disagreements.

☐ Maintain decision records, changelog entries, setup instructions, and reproducible environment files.

☐ Reproduce final runs from a clean checkout and prepare the application demonstration, report, and presentation.

13 Definition of Done

1. **Agentic dashboard prototype:** support criterion search, ranked cross-protocol retrieval, side-by-side source and structured differences, LLM evidence and rationale, exact checks, agent trace, uncertainty, review controls, abstention/escalation, export, and audit records.
2. **Versioned benchmark:** include source context, identifiers and versions, relationship labels, evidence, rationales, adjudications, anchor pairs/triplets, sampling provenance, reuse conditions, and independent trial-family splits.
3. **Baseline gap report and training artifacts:** document lexical and encoder failures, provide the reproducible fine-tuning pipeline and trained open encoder, and report supervision and component ablations.
4. **Held-out test report:** cover embedding improvement, equivalence retrieval, relationship classification, LLM quality, reviewer effort, agent reliability, calibration, abstention, latency, cost, and clinical failure analysis.
5. **Reproducibility package:** include permitted data or a complete acquisition manifest, annotation rules, references, prompts, model and dependency versions, model formatting, random seeds, and instructions sufficient for another team to run the prototype.

Final test answers will remain separate from training artifacts and agent retrieval. Public release will include only data and source material whose reuse conditions permit distribution; otherwise, the manifest and reconstruction instructions will be provided.

14 Implementation Schedule

Dates	Goal
October 3–6	Everyone reads AutoCriteria. Agree on relationship labels, therapeutic areas, protocol selection, annotation format, and software interfaces.
October 7–12	Run Modern AutoCrit on the first protocols. Test frozen embedding models. Retrieve nearest neighbors and collect clear failure examples. Pilot the LLM labeler and human-review form.
October 13–16	Label a small multi-domain dataset, measure initial reviewer agreement, build the first training dataset, and implement basic agent checks.
October 17–19	Produce preliminary retrieval results, failure examples, a working agent trace, and, if enough labels exist, a small fine-tuning experiment. Prepare the presentation.
October 20	**Progress presentation:** explain AutoCriteria and Modern AutoCrit; describe the corpus; show embedding failures; explain the LLM-to-human labeling loop; report baseline results; demonstrate one agent run; present risks and the remaining plan.
October 21–November 2	Expand protocols and labels, improve prompts and checks, finish double review, collect manuscript summaries, and freeze train/validation/test splits.
November 3–16	Fine-tune the encoder, build the relationship classifier, calibrate it, and run the main ablations.
November 17–30	Run held-out tests, agent and abstention experiments, manuscript population analysis, robustness checks, and error analysis.
Final period	Freeze data and models, reproduce results, finish the application, write the report, and prepare the final presentation.

15 Engineering Backlog

Before October 20

☐ Read the AutoCriteria paper and review the Modern AutoCrit input/output.

☐ Choose one or two therapeutic areas and define which protocols qualify.

☐ Agree on relationship labels and write short examples for each label.

☐ Run Modern AutoCrit on the first protocol batch and inspect its output.

☐ Define the path toward 80–120 trials and approximately 2,000 reviewed atomic criteria, while documenting the smaller progress-presentation subset honestly.

☐ Select and record three frozen generic encoders, one biomedical encoder if feasible, and one open checkpoint eligible for fine-tuning.

☐ Retrieve nearest neighbors and collect multi-domain failure examples.

☐ Build the LLM label format, rule checks, and human-review form.

☐ Double-label a pilot dataset, calculate agreement, and define qualified adjudication for unresolved cases.

☐ Split data by trial family before pairing; group amendments, templates, near duplicates, and constructed derivatives and check for leakage.

☐ Create a minimal fine-tuning dataset and training script.

☐ Demonstrate one full agent and review workflow.

☐ Prepare the October 20 slides, figures, and risk list.

After October 20

☐ Expand labels using uncertainty and tool disagreement.

☐ Define positive, ordinary-negative, and hard-negative sampling.

☐ Fine-tune and calibrate the dense encoder.

☐ Train the relationship classifier.

☐ Test the contribution of hard negatives, LLM suggestions, human corrections, and the critic.

☐ Review aggregate population-comparison literature and define allowed claims.

☐ Extract comparable manuscript baseline characteristics with provenance.

☐ Evaluate retrieval, relationship classification, calibration, and agent routing.

☐ Compare uncertainty/disagreement sampling with random sampling under a matched review budget.

☐ Run blinded test annotation without model or LLM suggestions and keep final labels inaccessible to agent retrieval.

☐ Report natural and constructed results separately, trial-family resampling uncertainty, and training-seed variation.

☐ Complete domain-specific and out-of-distribution error analysis.

☐ Freeze code, prompts, data, model versions, and reproducibility instructions.

16 Literature Worth Reviewing

The goal of this reading list is to resolve implementation choices, not to accumulate background citations. For each paper or resource, add a short note under `docs/references/` recording the problem, data, supervision, split strategy, metrics, reusable code or model, and one design decision it changes for TriAlign.

Eligibility extraction and representation

- **AutoCriteria** establishes the extraction problem and the system that Modern AutoCrit extends [1]. Review its output representation, evaluation design, and failure modes before defining the TriAlign adapter.

- **Chia** provides an expert-annotated eligibility corpus and a useful semantic decomposition [3]. Review it for external terminology and evaluation ideas even if it is not included in the primary training corpus.
- **Modern AutoCrit** is the operational upstream contract [2]. Review the validated export schema, identifier rules, review semantics, and provenance fields rather than treating the spreadsheet as anonymous text.

Sentence embeddings and representation learning

- **Sentence-BERT** explains bi-encoder sentence representations and efficient cosine retrieval [6]; it is the architectural starting point for the frozen and fine-tuned encoders.
- **SimCSE** is relevant to contrastive objectives, negative construction, and representation quality [7].
- **BGE/C-Pack** is relevant to retrieval-oriented encoders and instruction-aware input formatting [8].
- **SapBERT** and **BioLORD** should be reviewed to understand the difference between biomedical concept alignment and complete-sentence eligibility meaning [9, 10].
- The current Sentence Transformers training documentation should be used for runnable loss, trainer, and evaluator patterns: `https://www.sbert.net/docs/sentence_transformer/training_overview.html`.

Clinical-trial retrieval and similarity

- **Trial2Vec** provides a direct clinical-trial document retrieval comparison and a warning that whole-trial similarity is not the same as criterion-level equivalence [5].
- Eligibility-cluster work should be reviewed for candidate mining and domain grouping, while checking whether its notion of similarity preserves operators, negation, and clinical direction [4].

Human labels, active learning, and agent evaluation

- Review active-learning literature on uncertainty, diversity, and disagreement sampling before claiming that agent-selected examples reduce annotation cost; compare every strategy with matched random sampling.
- Review inter-rater reliability and adjudication guidance before selecting Cohen's κ, weighted κ, Krippendorff's α, or prevalence-aware alternatives. Agreement must be reported by relationship class, not only as one overall number.
- Review LLM-as-a-judge limitations and position/order bias before designing the critic. Pair reversal, evidence support, abstention, and blinded human labels should be explicit tests rather than prompt assumptions.

Calibration, abstention, and statistical evaluation

- **On Calibration of Modern Neural Networks** motivates post-hoc calibration and expected calibration error, including their limitations [11].

- Selective-prediction and risk–coverage literature should guide abstention thresholds and escalation evaluation rather than using confidence as an unexplained score [12].
- Clustered resampling literature should be reviewed before implementing confidence intervals because pairs sharing a trial family are not independent.
- Population-adjustment literature remains relevant only to the optional manuscript analysis and its limits [13, 14].

17 Scope Boundaries

The minimum complete project is a one- or two-area protocol corpus, lexical matching, three frozen general encoders, a biomedical baseline where feasible, a reviewed pair and anchor dataset, one fine-tuned open dense encoder, a relationship classifier, one agentic labeling/review workflow, and leakage-controlled held-out evaluation. The target dataset is 80–120 trials and approximately 2,000 atomic criteria; achieved counts and exclusions will be reported transparently.

The manuscript baseline-characteristics analysis is included only if enough comparable variables are available and the literature review supports the chosen summaries. We will not claim individual-level population disagreement, automated patient eligibility, causal transportability, clinical risk, or simulated patient outcomes without suitable data. We will also avoid an unexplained weighted score that mixes unrelated quantities.

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